4 February 1987
SUPERSEDING
MIL-U-17611F
28 March 1979

MILITARY SPECIFICATION

# UNDERSHIRT, EXTREME COLD WEATHER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for a man's cold weather cotton knitted undershirt.

1.2 Classification. The undershirt shall be of one type and in the following sizes (see 6.2):

Sizes - X-Small, Small, Medium, Large, X-Large and XX-Large.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

ASMC N/A FSC 8415

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## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| V-T-276 | - Thread, Cotton |
| NN-P-71 | - Pallets, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial) |
| DDD-L-20 | - Label; for Clothing, Equipage and Tentage (General Use) |

MILITARY

| | |
|---|---|
| MIL-P-15011 | - Pallets, Material Handling, Wood, Post Construction, 4 Way Entry |
| MIL-C-17157 | - Cloth, Knitted, Cotton (Waffle Type) |
| MIL-B-17757 | - Boxes, Shipping, Fiberboard (Modular Sizes) |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-191 | - Textile Test Methods |
| FED-STD-751 | - Stitches, Seams, and Stitchings |

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | - Marking for Shipment and Storage |
| MIL-STD-147 | - Palletized Unit Load |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall bse those in effect on the date of the solicitation.



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LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington D.C. 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of documents which are DOD adopted shall be those listed in the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606).

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material. (See 6.4)

3.3.1 Cloth, knitted, cotton (waffle type). The basic material shall conform to MIL-C-17157, either flat knit or circular knit.



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3.3.2 Cloth, cotton, rib-knit. The cloth for the wristlets and collarette shall be knitted from unbleached cotton and shall conform to the following requirements when tested as specified in Table IV:

| Knit | Wales and courses per inch (2.5 cm)(min) | | Yarn size | Bursting strength (min) lbs. (newtons) |
|---|---|---|---|---|
| | Wales | Courses | | |
| 1 by 1 rib | 20 | 35 | 18/1 | 95 (423) |

* 3.3.3 Thread, Cotton. The thread for seaming and stitching shall be unbleached, type IA1, Ticket Nos. 50 and 70, 3 ply conforming to V-T-276.

3.3.4 Yarn, coverstitching. The yarn for coverstitching the seams shall be unbleached cotton in sizes 28/1 (lower) and 30/2 (upper) and shall conform to the following requirements when tested as specified in Table IV:

| Yarn sizes | Lengths (min) | | Breaking strength (min) | |
|---|---|---|---|---|
| | Yards/lbs. | Meter/kg | Pounds | Newtons |
| 28/1 1/ | 21,000 | 42,185 | 0.50 | 2.23 |
| 30/2 | 11,000 | 22,097 | 1.65 | 7.34 |

1/ In lieu of 28/1 sewing yarn, a ticket No. 90, type IA1, 2 ply thread conforming to V-T-276 may be used as an alternate on coverstitching the seam.

3.3.5 Label. The undershirts shall have a combination identification, size and instruction label conforming to type VI, class 14 of DDD-L-20. The label printing shall show fastness to laundering. The label and format shall be as follows:

UNDERSHIRT, EXTREME COLD WEATHER
FIBER CONTENT: 100% COTTON
CONTRACT NO.: DLA100-00-0-0000 (EXAMPLE)
STOCK NO.: 8415-00-00-0000 (EXAMPLE)
SIZE: MEDIUM (EXAMPLE)
NAME OF CONTRACTOR:

LAUNDERING INSTRUCTIONS
Hand or Machine wash - hot water
Tumble dry or flat dry
DO NOT HANG TO DRY



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* 3.4 Design. The undershirt shall be a pullover type having long sleeves, wristlets and a collarette (See figure 1). When circular knit fabric is used, one or both sides may be seamless.

3.4.1 Figure. The figure is furnished for information purposes only. To the extent of any inconsistencies between the specification and the figure, the specification shall govern.

* 3.5 Patterns. Standard patterns will be furnished by the Government to the contractor for use in cutting the working patterns (see 6.3). The Government patterns shall not be altered in any way and the working patterns shall be identical to the Government patterns.

3.5.1 List of pattern parts. The component parts for the undershirts shall be cut from materials as specified below:

| Material | Pattern Nomenclature | Cut Parts |
|---|---|---|
| Cloth, Cotton, Knit (waffle type) | Back 1/ | 1 |
| | Front 1/ | 1 |
| | Sleeve | 2 |
| Cloth, cotton, rib-knit | Collarette | 1 |
| | Wristlet | 2 |

1/ There is no seam allowance added for flat knit vs circular knit since the seam for joining the sides is FSa-1 which is an abutted seam. When using circular knit goods, the patterns are used to insure that the measurements obtained are the same as those obtained if flat knit goods were used.

3.6 Construction.

* 3.6.1 Stitches, seams and stitchings. Stitch, seam and stitching types specified in Table I shall conform to FED-STD-751. Whenever two or more methods for seam or stitch types are given for the same part of the operation any one may be used. Seam allowance shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers or open seams result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching. Where stitch types 604, 605, 606 or 607 are used, the underthread shall be on the inside of the undershirt.



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* 3.6.2 Thread breaks and end of seams. Ends of all seams and stitching when not caught in other seams or stitching shall be backtacked not less than 1/2 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks or skipped stitches (all stitch types) shall be repaired by stitching back of the break not less than 1/2 inch. When ends of overedge stitching are not caught in other seams or stitching, the ends shall be tacked.

3.6.3 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table I.

3.7 Manufacturing operations requirements. The undershirt shall be manufactured in accordance with the operations specified in Table I. The contractor is not required to follow the exact sequence of operations as specified, provided the finished undershirts are identical with those produced by following the sequence as specified in Table I.

* 3.7.1 Abbreviations in table of operations. The abbeviations used in table I are as follows:

| | | |
|---|---|---|
| Stch | - | Stitch |
| in | - | inch |
| Ndl | - | Needle |
| Bob | - | Bobbin |
| Lpr | - | Looper |
| Mchne | - | Machine |
| Brtck | - | Bartack |
| Comrcl | - | Commercial |
| smlr | - | similar |
| Btnhl | - | Buttonhole |
| incl | - | including |
| dbl | - | double |



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TABLE I - CONSTRUCTION OF UNDERSHIRT

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| 1. | Cutting of undershirts.<br>a. Due to the distinctive stretch properties of this type knit fabric and in order to meet the finished garment measurements specified in Table II, it is recommended that the fabric be allowed to relax for at least 12 hours before cutting.<br><br>b. All parts of the undershirts shall be cut in strict accordance with patterns furnished which show size, shape, directional lines and notches for the proper assembly. The directional lines may vary from the wale direction by not more than 2-1/2 inches. This deviation shall be checked by measuring from the top and bottom of the directional line on pattern to the edge of fabric. The difference between the two measurements shall not exceed the tolerance specified. | | | | | |
| 2. | Replacement of defective components.<br>During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in Section 4 shall be removed from production and replaced with non-defective and properly matched components. | | | | | |
| 3. | Shoulders and collarette.<br>a. Close the right shoulder. | 606 or 607 | FSa-1 | 12-14 | 70 | 30<br>28 |
| | b. Collarette shall be double thickness of ribbed knit cloth. | | | | | |
| | c. Join the folded collarette to neckline of the undershirt, starting and finishing at the opening of the left shoulder with the undershirt material inserted approximately 5/8 inch. The finished width shall measure 1-1/4 (±1/8) inches with the wales of the collarette material perpendicular to the collarette seam. | 604 or 605 | BSa-1 | 12-14 | 50 | 50<br>50 |



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TABLE I - CONSTRUCTION OF UNDERSHIRT

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| | d. Close the left shoulder and collarette. | 606 or 607 | FSa-1 | 12-14 | 70 | 30<br>28 |
| | e. Turn the stitching chain at the end of collarette to the inside and secure with a 1/2 or 5/8 inch bartack, either centered across or parallel to the left shoulder seam, 1/8 to 1/4 inch from the folded edge of the collarette. | Brtck | | 28 (1/2 in) or 42 (5/8 in) per brtck | | |
| 4. | Wristlets.<br>a. Wristlets shall be double thickness of ribbed knit cloth. | | | | | |
| | b. Lap the folded rib-knit cloth over the bottom edge of each sleeve 1/4 to 3/8 inch and stitch with the wales of the rib-knit cloth perpendicular to the wristlet seam. | 604 or 605 | LSa-1 | 12-14 | 50 | 50<br>50 |
| 5. | Armholes.<br>a. Set sleeves to each armhole. | 606 or 607 | FSa-1 | 12-14 | 70 | 30<br>28 |
| 6. | Assembly.<br>a. Close side seam, sleeve inseams and wristlet in a continuous seaming operation for each side. | 606 or 607 | FSa-1 | 12-14 | 70 | 30<br>28 |
| | Note: As an alternate, the side seam and sleeve inseam may be closed separately prior to insertion of sleeve into armhole provided the two seams are aligned at underarm and the final seam is secured by overstitching. Also, delete all reference to side seam closing when circular knit material is used for the body. | | | | | |
| | b. Turn the stitching chain at the end of the wristlet to the inside and secure with a 1/2 or 5/8 inch bartack, either centered across or parallel to the seam, 1/8 to 1/4 inch from the bottom edge of wristlet. | Brtck | | 28 (1/2 in ) or 42 (5/8 in ) per brtck | | |



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TABLE I - CONSTRUCTION OF UNDERSHIRT

| No. | Operation | Stch Type | Seam/ Stch type | Stch in | Thread Ndl | Thread Bob/ Lpr |
|---|---|---|---|---|---|---|
| 7. | Bottom edge.<br>a. Overedge bottom of the undershirt. | 504 or 505 | EFd-1 | 12-14 | 50 or 70 | 50 or 70 |
| 8. | Label.<br>a. Center the garment label on the inside back of the undershirt, with the top of the label 1/2 (+1/4) inch below the collarette seam. Stitch on all four sides 1/16 to 1/8 inch from the edge. | 301 | LSbj-1 | 12-14 | 50 | 50 |
| 9. | Clean undershirt.<br>a. All thread ends shall be trimmed and loose threads removed from the undershirt.<br>Note: Caution shall be made so not to cut the material when trimming.<br>b. Remove all spots and stains. | | | | | |



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* 3.8 Sizes and measurements. Sizes and measurements shall conform to the requirements in Table II. All measurements and tolerances are expressed in inches and shall be taken with undershirt laid out flat and smooth, face up and under no tension.

Table II - Sizes and measurements

| Size | Chest (A) | Length (B) | Sleeve length (C) | Shoulder length (D) | Wristlet length (E) | Wristlet width (F) |
|---|---|---|---|---|---|---|
| X-Small | 12-1/2 | 24 | 23-1/4 | 2-3/4 | 3-1/2 | 3-3/4 |
| Small | 14-1/2 | 25 | 23-1/2 | 3-1/4 | 3-1/2 | 3-3/4 |
| Medium | 16-1/2 | 26 | 23-1/2 | 3-3/4 | 3-1/2 | 3-3/4 |
| Large | 18-1/2 | 27 | 23-3/4 | 4-1/4 | 3-1/2 | 3-3/4 |
| X-Large | 20-1/2 | 28 | 23-3/4 | 4-3/4 | 3-1/2 | 3-3/4 |
| XX-Large | 22-1/2 | 29 | 24 | 5 | 3-1/2 | 3-3/4 |
| Tolerance | ± 1 | ± 1 | ± 1 | ± 1/2 | ± 1/4 | ± 1/4 |

NOTE: Refer to Figure 1 for A, B, C, D, E and F measurements.

A. Measure across front, approximately 1/2 inch below the intersection of the armhole, from side seam to side seam or with circular knits from folded edge to folded edge.

B. Measure from the intersection of the collarette seam and the shoulder seam in a line perpendicular to the bottom of the undershirt.

C. Measure along the outside folded edge of sleeve from the intersection of the armhole seam and the shoulder seam to the bottom of the wristlet.

D. Measure along the shoulder seam from the intersection of the shoulder seam and the collarette seam to the intersection of the shoulder seam and the armhole seam.

E. Measure from the top edge of seam to bottom folded edge at midpoint.

F. Measure from folded edge to folded edge at midpoint.

3.9 Workmanship. The finished undershirts shall conform to the quality established by this specification. The occurrence of defect shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS



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4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2, shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, appearance and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4.1.1 Testing of components. In addition, components and materials listed in Table IV shall be tested for the characteristics specified and in accordance with the referenced test methods in FED-STD-191. All test reports shall contain the individual values utilized in expressing the final results. The sample size shall be in accordance with Table III and the lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample unit and basis for lot size shall be expressed as follows:



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| Material | Basis for lot size | Sampling unit for testing |
|---|---|---|
| Cloth, cotton rib-knit | Pound | 1 yard full width |
| Yarn, cotton | Spool, cone or tube | 1 holder |

Table III - Sample size

| Lot size | Sample size |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |

Table IV - Component testing

| Material | Characteristic | Requirement paragraph | Test method |
|---|---|---|---|
| Cloth, Cotton, rib-knit (Wristlet and collarette) | Cotton identification | 3.3.2 | 1200 1/ |
| | Yarn size | 3.3.2 | 4010 1/ |
| | Knit type | 3.3.2 | Visual 2/ |
| | Wales and courses/inch | 3.3.2 | 5070 |
| | Bursting strength | 3.3.2 | 5120 |
| Yarn, cotton | Cotton identification | 3.3.4 | 1200 1/ |
| | Yarn size | 3.3.4 | 4020 |
| | Yards per lbs. | 3.3.4 | 4010 |
| | Breaking strength | 3.3.4 | 4100 |

1/ Unless otherwise specified, a contractor's certificate of compliance shall be acceptable for these characteristics.
2/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

4.4.2 Examination of the end item. The defects found during the examination of the end item shall be classified in accordance with 4.4.2.1 and 4.4.2.2. The applicable inspection levels and acceptable quality levels (AQL'S) shall be as indicated in 4.4.2.3. The sample unit shall be one undershirt. The lot size shall be expressed in units of undershirts.



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4.4.2.1 Visual examination. The undershirts shall be visually examined for the following defects:

| Defect | Classification Major | Minor |
|---|---|---|
| I MATERIAL DEFECTS AND DAMAGES | | |
| a. Any hole, cut, tear, drop stitch, needle chew or mend or thin area. | X | |
| b. Slubby yarn, more than twice the size of the normal yarn | | X |
| c. Any loose knitting, causing sleaziness | | X |
| d. Pulled or snagged stitches | | X |
| II CLEANLINESS | | |
| a. Any spot or stain | | X |
| b. Foreign matter | | X |
| c. Objectionable odor | X | |
| d. Five or more ends not trimmed to less than 1/2 inch inside. | | X |
| e. Three or more ends not trimmed to less than 1/4 inch on the outside. | | X |
| III COMPONENTS AND ASSEMBLY | | |
| a. Collarette improperly attached | X | |
| b. Any component part or operation omitted, not as as specified, or improperly performed. | X | |
| c. Any clearly visible distortions or area of tension | X | |
| IV CUTTING | | |
| a. Any component part not cut in accordance with specified pattern, directional lines on pattern or not in accordance with specification requirements. | X | |
| V SEAMS AND STITCHING | | |
| a. Accuracy of seaming. | | |
| 1. Any seam twisted, puckered or pulled | | X |
| 2. Loose or tight tension resulting in an unsatisfactory seam. | X | |
| 3. Part of garment caught in any unrelated operation or stitching. | X | |
| 4. End of seam or stitching, when not caught in other seam or stitching, tacked or backtack less than 1/2 inch. | | X |
| 5. Stitching overlapping ends of thread breaks less than 1/2 inch. | | X |



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| Defect | Classification Major | Minor |
|---|---|---|
| b. Open seams<br>Any open seams. | X | |
| Note: A broken stitch or two or more continuous stitches constitutes an open seam. | | |
| c. Raw edge<br>Any protrusion of more than 1/4 inch from stitching line or protruding more than 1/8 inch along one inch or more of seam. | | X |
| d. Runoff - More than 1/4 inch | | X |
| e. Not specified seam or stitch type. | X | |
| f. Number of stitches per inch is more or less than specified. | | X |
| g. Bartacks- | | |
| 1. Missing, insecure or misplaced | | X |
| 2. Length less than 1/2 inch or more than 5/8 inch. | | X |
| VI LABEL | | |
| a. Missing, misplaced, incorrect or illegible | | X |
| b. Stitched improperly | | X |

* 4.4.2.2 Dimensional examination. The undershirts shall be examined for conformance to the measurements specified in Tables II. Measurements will be made as described in footnotes below the table. Any measurements not within the specified tolerances shall be scored as a defect.

4.4.2.3 Acceptable quality and inspection levels. The acceptable quality levels (AQL'S) expressed in defects per hundred units and the inspection levels shall be as follows:

| Visual examination (4.4.2.1) | AQL'S | Inspection levels |
|---|---|---|
| Major defects | 2.5 | II |
| Total defects (Major and Minor combined) | 6.5 | II |
| Dimensional examination (4.4.2.2) | | |
| One class of defects | 4.0 | S-3 |



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4.5 Examination of packaging requirements. An examination shall be made to determine that packaging, packing and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

| Examine | Defects |
|---|---|
| Marking (interior and exterior) | Omitted, incorrect, illegible, of improper size, location, sequence, or method of application. |
| Material | Any component missing, damaged or not as specified. |
| Workmanship | Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, inadequate stapling. Distorted or bulging containers. |
| Content | Number of items per shipping container is more or less than specifed. 1/ Size shown on one or more undershirts not as specified on container. 1/ |

1/ For these defects, two shipping containers shall be examined from each lot.

*4.6 Palletization examination. An examination shall be made to determine that the palletization complies with the Section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 as specified in MIL-STD-105.

| Examine | Defect |
|---|---|
| Finished dimension | Length, width, or height exceeds specified maximum requirements. |
| Palletization | Pallet pattern not as specified.<br>Interlocking of loads not as specified.<br>Load not bonded with required straps as specified. |
| Weight | Exceeds maximum load limits. |
| Marking | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application. |



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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each undershirt shall be laid face down, each side shall be folded in 2 to 4 inches (according to size) with the sleeves parallel to the sides, then folded in half by bringing bottom of garment up to the top. Each garment shall measure approximately 11-1/2 inches in width by 14-1/2 inches in length. Five folded undershirts on one size only, with every other one reversed shall be neatly stacked and securely tied across the center width with twine or tape to form a bundle.

5.1.2 Level C. The undershirts shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Level A. Thirty (30) undershirts, of one size only, packaged as specified in 5.1 shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class weather-resistant, grade V15c, variety DW, size 3A of MIL-B-17757. The fiberboard for the liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages shall be packed flat in two stacks, three bundles per stack within the shipping container. Each container shall have the contents completly covered on the top and bottom with a sheet of commerical grade kraft paper. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

* 5.2.2 Level B. Thirty (30) undershirts, of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class domestic, grade 275, variety DW, size 3A of MIL-B-17757. The fiberboard for the liner shall conform to type CF, class domestic of MIL-B-17757. Level A packages shall be packed flat in two stacks, three bundles per stack within the shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commerical grade kraft paper. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.3 Level C. The undershirts, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classificaton Rules as applicable.



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5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.1 Label, mixed sizes. Each shipping container, packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a 4 by 5 inch white paper label with the words "MIXED NSN" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and National Stock Numbers contained therein.

* 5.4 Palletization. When specified (see 6.2), item, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be banded with primary and secondary straps in accordance with the banding means C, K and L, or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, Type IV; Type V, class 1, size 2; or Type VIII fabricated from wood group I, II, III or IV, Grade A of NN-P-71, or 4-way, Style 1, Size A, Type I, Class 1 fabricated from wood groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

6. NOTES

6.1 Intended use. The undershirts covered by this specification are intended for wear by male personnel as part of the cold weather outfit.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size required.
(c) When first article inspection is required (see 3.2), the item will be tested and should be a first article sample. The contracting officer shall include specific instructions in acquisition documents regarding arrangement for examinations, quantity, and testing and approval of the first article.
(d) Selection for applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When palletization is required (see 5.4).

6.3 Samples and patterns. For access to samples and patterns, address the procuring office issuing the invitation for bids.

6.4 Recycled material. It is encourged that recycled material be used when practical as long as it meets the requirements of this specification.



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* 6.5 Subject term (key word) listing.

Clothing, cold weather
Undershirt, waffle knit
Undershirt, 100% cotton
Undershirt, thermo
Underwear, thermo
Undershirt, unisex

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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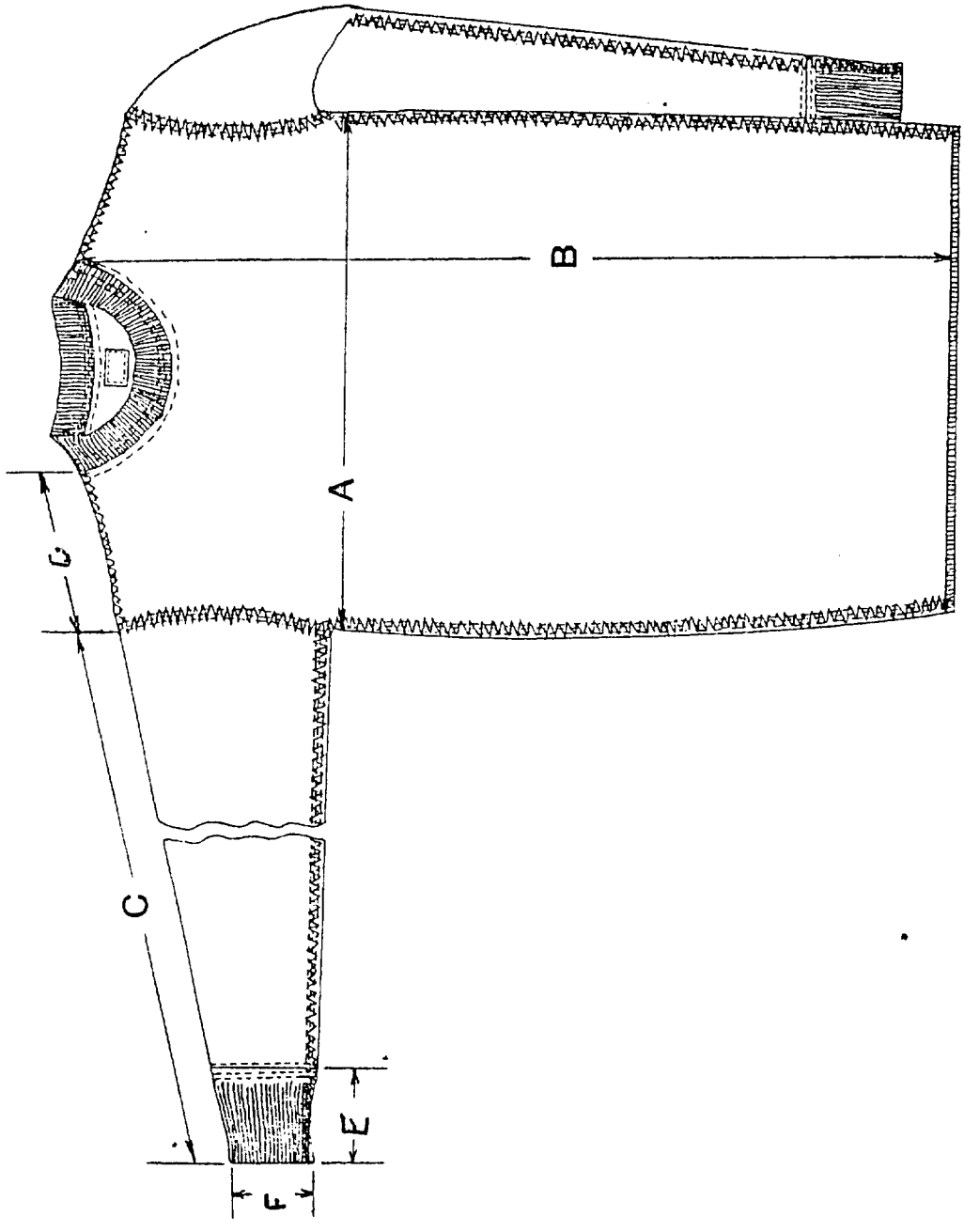



FIGURE 1 - UNDERSHIRT; EXTREME COLD WEATHER

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-U-17611G | UNDERSHIRT, EXTREME COLD WEATHER |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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